Hai-Long Qin

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Biography

Hai-Long Qin is a master's student at the School of Artificial Intelligence, Beijing University of Posts and Telecommunications (BUPT). His research interests lie in **Machine Intelligence** and **Cognitive Communication**, with a current focus on **Generative World Modeling**. He aims to develop generative world models that allow machines to understand, convey, and interact with human intent more efficiently.

Education

Beijing University of Posts and Telecommunications (BUPT) **Sep. 2024 – Present**
M.S. in Artificial Intelligence (Intelligent Information Processing) *Beijing, China*

- Key Laboratory of Universal Wireless Communications (Ministry of Education).
- School of Artificial Intelligence, admitted without the entrance examination.

Jiangsu University **Sep. 2020 – Jun. 2024**
B.S. in Communication Engineering, with a minor in Data Science *Zhenjiang, China*

- Future Network Laboratory, School of Computer Science and Communication Engineering.
- Elite College of Jiangsu University (21st-Century Outstanding Talent Class), with interdisciplinary learning.

Selected Publications

*[J]: Journal, [C]: Conference; **Bold**: my name, †: corresponding author, *: equal contribution.*

[J] Generative AI Meets 6G and Beyond: Diffusion Models for Semantic Communications
Hai-Long Qin, Jincheng Dai†, Guo Lu, Shuo Shao, Sixian Wang, Tongda Xu, Wenjun Zhang, Ping Zhang, Khaled B. Letaief
IEEE Commun. Surv. Tutor. (COMST), 2026 (**CAS Q1**; **Impact Factor 46.7**; Early Access)

- **Contribution:** The first systematic tutorial on diffusion models for generative semantic communications. Starting from the score-matching foundation, it organizes three research threads (conditional diffusion for controllable generation, efficient diffusion for accelerated inference, and generalized diffusion for cross-domain adaptation), and formulates receiver-side semantic decoding as an inverse problem solved via diffusion posterior sampling. Across human-, machine-, and agent-centric semantic communication scenarios, it explains how diffusion models preserve visual fidelity and channel robustness at extremely low bitrates.

[C] Neural Hamiltonian Deformation Fields for Dynamic Scene Rendering
Hai-Long Qin, Sixian Wang, Guo Lu, Jincheng Dai†
SIGGRAPH Asia 2025, Hong Kong, China (**Top-tier computer graphics conference**; **Oral**)

- **Problem:** Existing dynamic Gaussian Splatting methods rely on purely data-driven MLPs to predict deformation fields, lacking physical constraints. This leads to discontinuous and physically implausible deformation trajectories, fails to properly separate spatial and temporal deformation, and produces primitive overlap and quality degradation in complex motion scenes.
- **Method:** Propose Neural Hamiltonian Deformation (NeHaD), replacing the MLP deformation decoder with a Hamiltonian neural network that learns energy conservation and related physical laws in phase space without supervision. Boltzmann equilibrium decomposition adaptively separates static and dynamic Gaussians by their deviation from equilibrium; second-order symplectic integration and local rigidity regularization ensure stable and realistic deformation. Scale-aware MipMapping and progressive LOD optimization further extend the method to bandwidth-constrained adaptive streaming rendering.
- **Contribution:** First to integrate Hamiltonian mechanics into neural Gaussian deformation, with physical constraints improving rendering realism and robustness. Extended to streaming rendering, achieving a better quality-efficiency trade-off on dynamic scenes.

[J] Neural Coding Is Not Always Semantic: Toward the Standardized Coding Workflow in Semantic Communications
Hai-Long Qin, Jincheng Dai†, Sixian Wang, Xiaoqi Qin, Shuo Shao, Kai Niu, Wenjun Xu, Ping Zhang†
IEEE Commun. Stand. Mag. (COMSTD), 2025 (**Semantic communication standardization**; **JCR Q1**; CiteScore 16.2)

- **Contribution:** First to address the foundational question of whether neural coding is equivalent to semantic coding. Proposes a standardized coding workflow for semantic communication that distinguishes general neural coding from true semantic coding, establishing that semantic communication is not merely feature transmission but the conveyance of compact semantic representations.

[J] Diffusion Model-Empowered Generative Visual Semantic Communication (in Chinese)
Hai-Long Qin, Jincheng Dai†, Sixian Wang, Shengshi Yao, Kai Niu, Wenjun Xu
Journal of Tsinghua University (Science and Technology), 2025 (**PKU Core Journal**; "Frontier Technologies for New-Quality Communications" Featured Paper; **Most Downloaded Article of 2025**)

[J] Intent-Driven Intelligent and Concise Communications (in Chinese)
Jincheng Dai, Xiaoqi Qin, **Hai-Long Qin**, Ping Zhang†
ZTE Technology Journal, 2025 (**PKU Core Journal**; "Time-Efficient Intelligent Machine Communication for 6G" Featured Paper)

Program Experience

National Training Program for Innovation and Entrepreneurship — **Sep. 2022 – Nov. 2023**
Project: Camera, LiDAR, and IMU Based Multi-Sensor Fusion SLAM for Indoor Robot Navigation — *Zhenjiang, China*
Team: **Hai-Long Qin**, Xingchen Zhang, Jingting Cai, Mengjie Zhang | Advisor: Prof. Yi Zhu

Summer Program for !nnovative Engineering Design (SP!ED 2023) — **Jul. – Aug. 2023**
Project: A Machine Vision-Based Automatic Waste Sorting Device — *Gimhae, South Korea*
Team: **Hai-Long Qin**, Shangmei Liu, Koki Kawata, Eunji Choi, Dongho Kim

Honors and Awards

***[N]**: National, **[P]**: Provincial.*

[N] National Scholarship for Graduate Students, 2025

[P] Excellent Undergraduate Thesis of Jiangsu Province, 2024
Thesis: *Research on Context-Aware Continuous Generative Modeling* | Advisors: Prof. Yi Zhu, Prof. Jincheng Dai

- **Problem:** Visual generative modeling suffers from normalization constraints, with limited flexibility, low quality, weak diversity, and high computational cost. Existing two-stage compress-then-generate schemes trade fidelity for performance without resolving these issues.
- **Method:** From a continuous distribution matching perspective, with Transformer as the contextual perceptor, propose three continuous generative models (H2ST, TDPS, HTAG) via sequence prediction, diffusion sampling, and adversarial learning, unified under Bayesian inference.
- **Contribution:** A deconvolution-dequantization-deunification paradigm that formulates image generation as a Bayesian probabilistic model guided by contextual semantic self-feedback. The three models outperform concurrent SOTA in FID on ImageNet and FFHQ.

[N] China University Intelligent Robot Creative Contest, 2022 & 2023 — **Silver & Bronze Awards**
Team: **Hai-Long Qin**, Xingchen Zhang, Yakai Dai, Qianchao Xu, Jingting Cai, Mengjie Zhang | Advisor: Prof. Yi Zhu

- Vision-based robot target detection and recognition, with lightweight on-device deployment of visual deep neural networks and low-latency communication.

[N] China Robot Competition & RoboCup China Open, 2022 — **Bronze Award**
Team: **Hai-Long Qin**, Xingchen Zhang | Advisor: Prof. Yi Zhu

- Multi-sensor (LiDAR, stereo cameras, IMU) fusion for simultaneous localization and mapping (SLAM), with path planning and high-precision navigation in open environments.

[N] China Collegiate Market Survey and Analysis Competition, 2023 — **Bronze Award**
Digital Economy Track | Team: Shihan Fan, Jingwen Ma, **Hai-Long Qin**, Tingya He, Na Wu | Advisors: Prof. Jun Yan, Prof. Aihua Gong

[N] China Collegiate Business Elite Challenge, 2022 — **Silver Award**
Innovation and Entrepreneurship Track | Team: Xinyu Gu, Min Xie, Jingwen Ma, Anbang Mao, **Hai-Long Qin** | Advisors: Prof. Yaya Li, Prof. Bai Lyu